

Development of Google Earth Engine-Based App for National Forest Inventory Plot Data Exploration

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Introduction

Efficient land use starts with understanding the available resources. Our team is focused on developing a comprehensive approach to land resource monitoring, leveraging both in-house and remotely sensed data to create a pipeline that generates land use and cover maps. Our objective is to provide land resource information to those who depend on the forests in Costa Rica.

1. Nature of the Data

- Nine land use classifications are determined and grouped per hectare. (Fig 1)
- These groups are called plots and are distributed in a standard grid across the landscape. (Fig 2)
- This sums up to 101,160 subplots from 11,240 plots, with data collected on use and cover types.

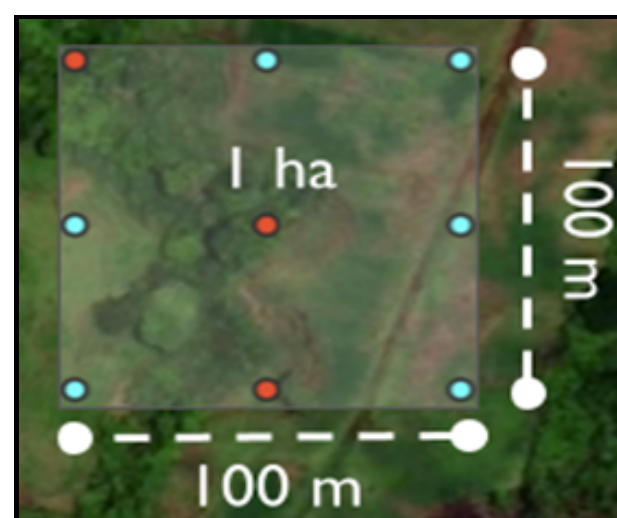


Fig 1. Example of one hectare plot with nine subplots

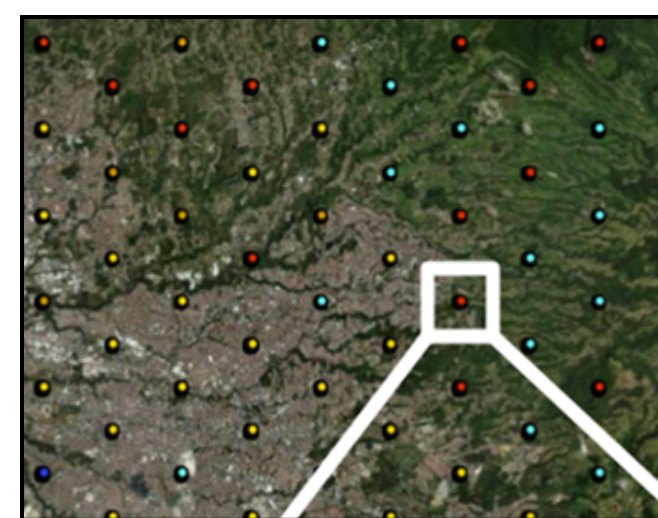


Fig 2. Example of Land Use Plots

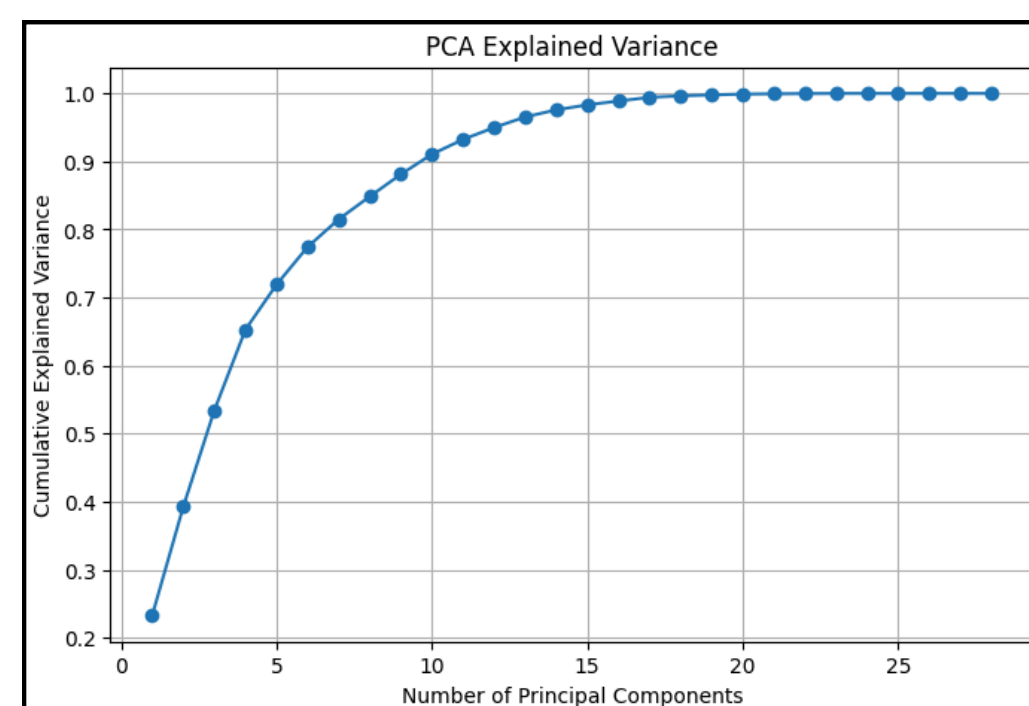
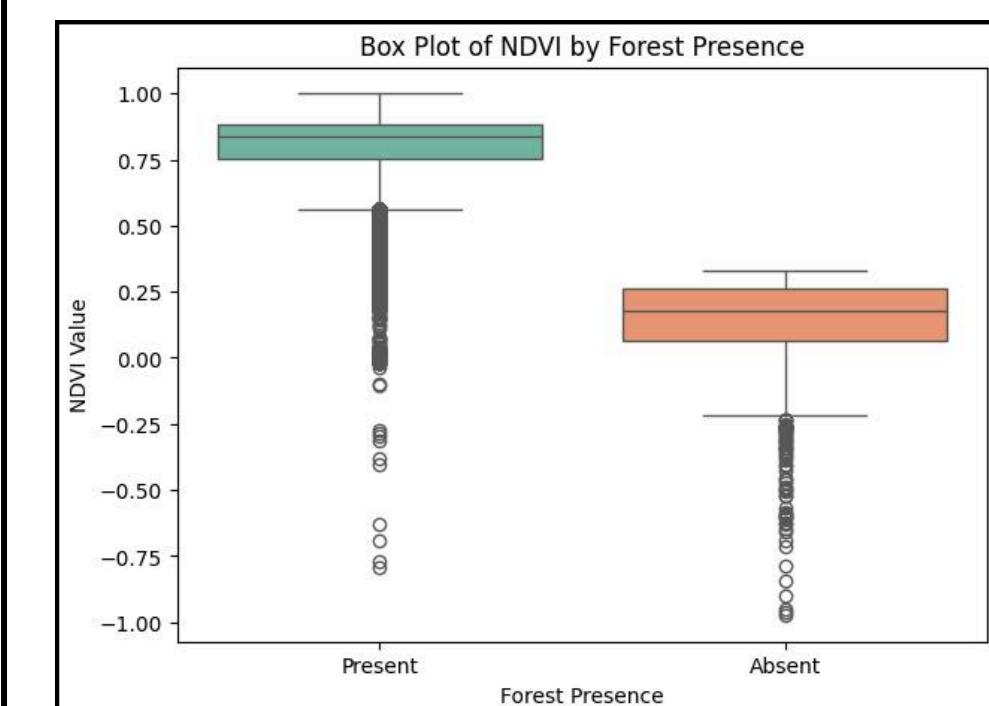
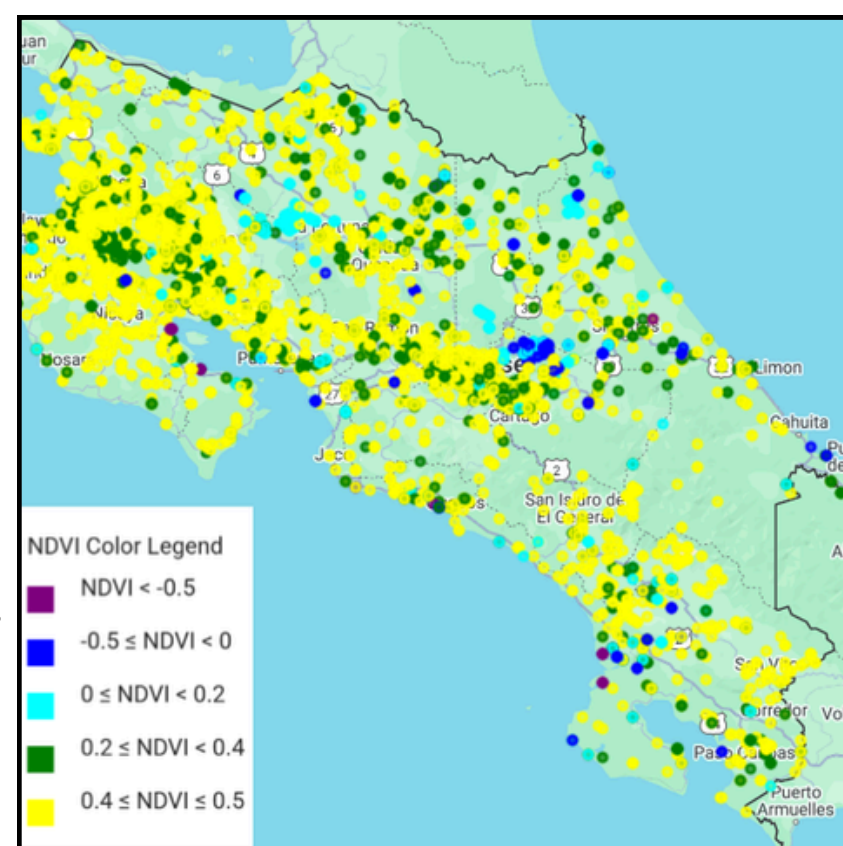
2. Exploratory Data Analysis (EDA)

Data Cleaning

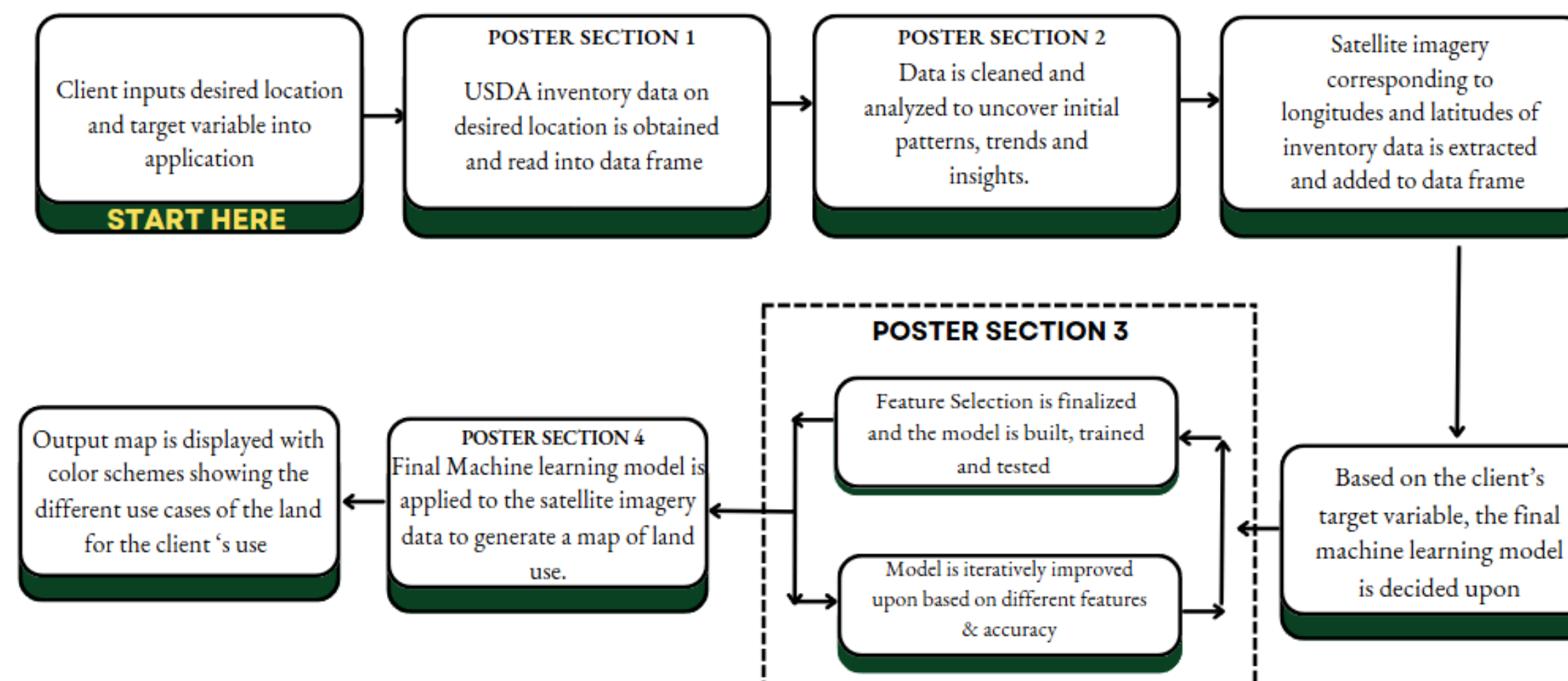
- Identified and removed quality assurance (QA) data
- Addressed missing (NA) instances
- Corrected mislabeled features and instances
- Translated Spanish diction to be English

Outlier Detection

- Identified and removed outlier points in dataset using box-and-whisker plots and other graphical analysis tools
 - This helped us characterize the distribution and relationships for each of the features



Product Workflow



4. App Dashboard

Dashboard Purpose:

- The application integrates remote sensing data with Random Forest models to classify vegetation types and assess forest conditions.
 - Cleans and filters spatial geometries for accuracy
 - Extracts key spectral features (e.g. NDVI, NIR, RED)
 - Trains and evaluates classification or regression models
 - Maps predictions using tools like Google Earth Engine and Folium
- The goal is to generate data driven maps to support land research and sustainable land management

Welcome to Our Interactive Geospatial Pipeline

We are a group of students from Purdue National Data Mine Network (NDMN) program, and our goal is to create an interactive pipeline that integrates pre-processing, machine learning models, and validation strategies for robust geospatial analysis, specifically tailored to Costa Rica's population. This project has been made possible with guidance from our mentors at the USDA Forest Service.

Step 1: Upload Raw Data

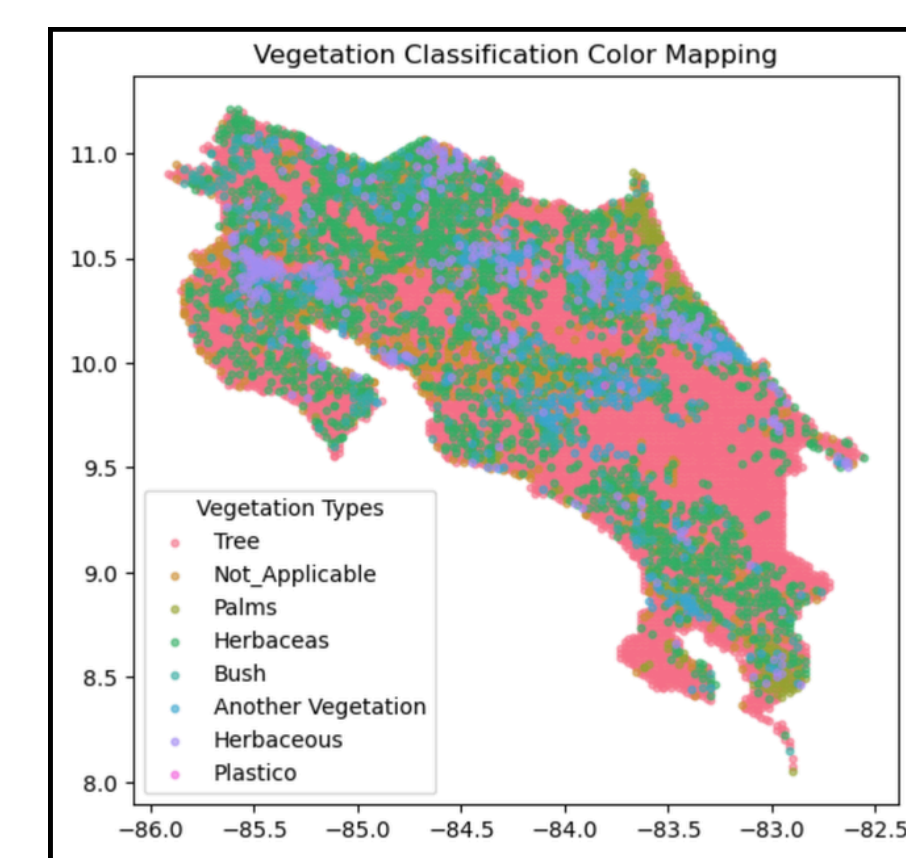
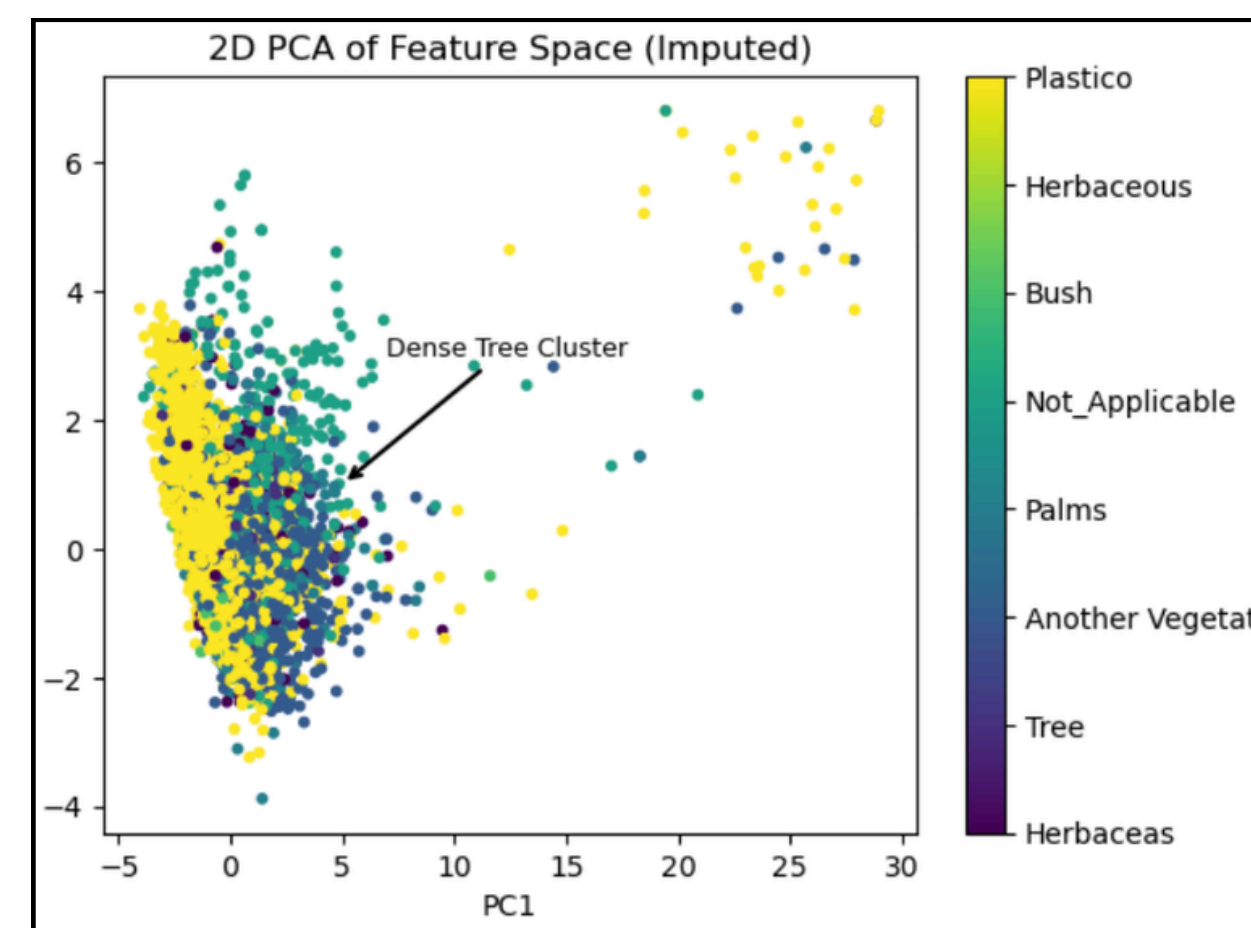
Upload your dataset
Drag and drop file here
Limit: 200MB per file • CSV, XLSX
Browse files
Costa_Classification_Data_Cleaned2.csv 11.2MB
Data uploaded successfully!

Step 2: Data Pre-Processing

Choose Pre-Processing Strategy
Remove Outliers
Pre-Processing: Remove Outliers

Step 3: Data Type Selection

What type of target variable?
Categorical
Continuous
Choose the attribute for continuous variable
% Forest

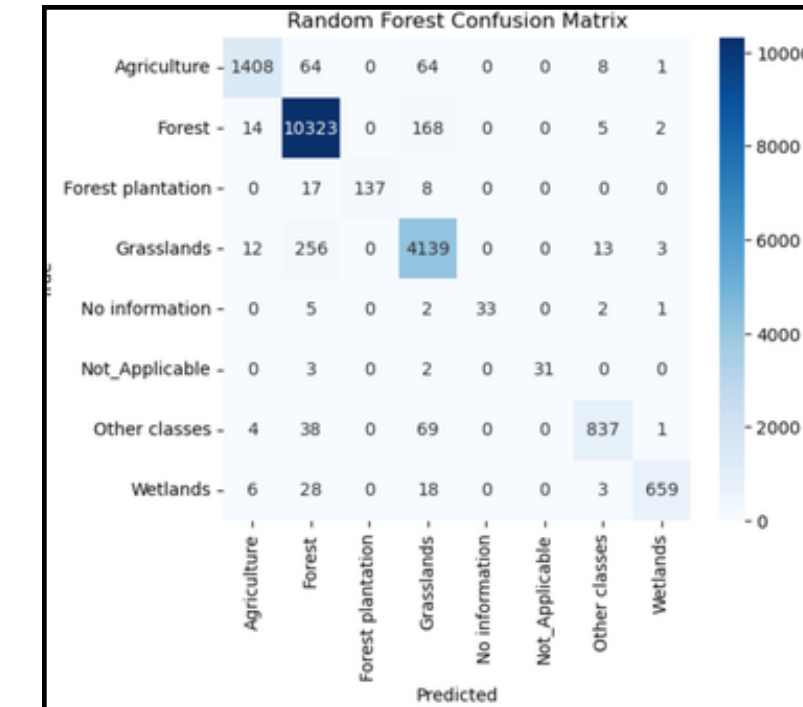


3. Machine Learning & Methodologies

Model & Feature Selection

Multiple Machine Learning (ML) classification models were trained:

- With land use as the response variable and geospatial data and satellite imagery as the predictor variables
- Logistic regression, Random Forest, Support Vector Machines were considered
- An 80-20 train-test split was used to partition the dataset into training and testing sets

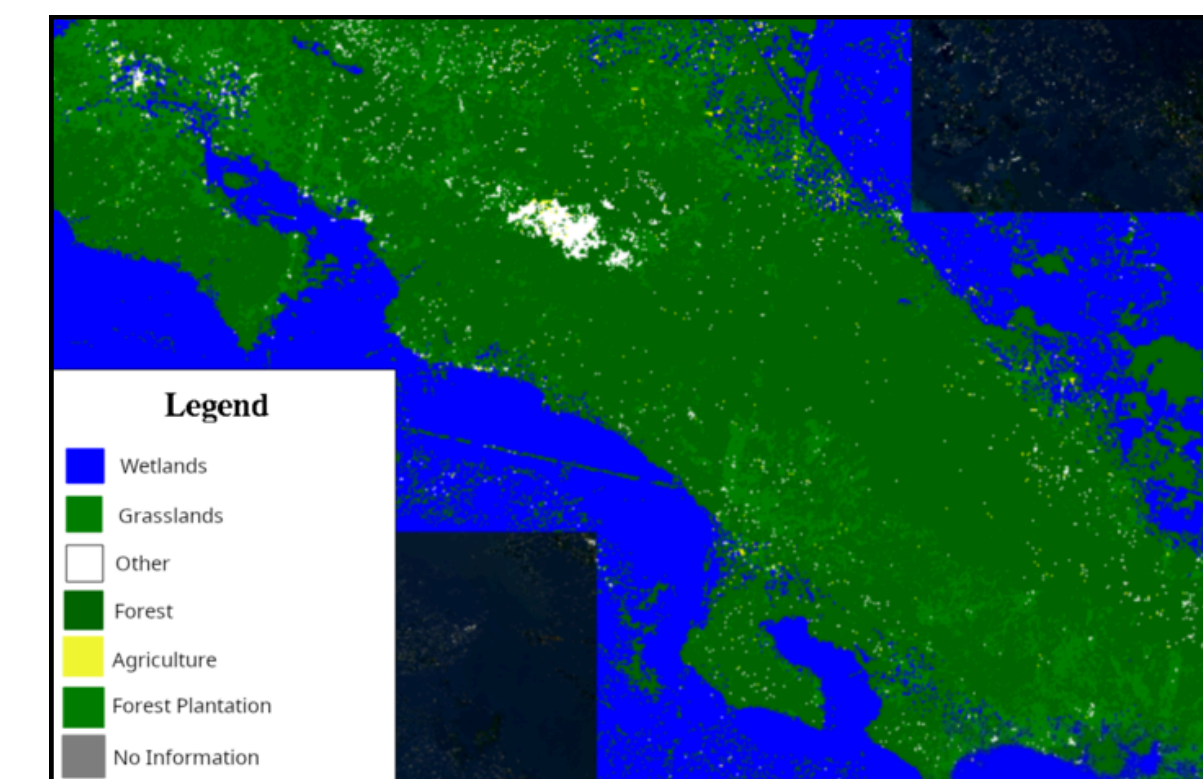


Model Results

- Confusion matrices were generated and models were evaluated by accuracy, precision, recall and F-1 score
- Random Forests performed the best at 70% accuracy
 - Results varied based on different test samples
- Determined that some of the most important features across models include elevation, NDVI, and other remote sensing variables such as red, green bands and shortwave infrared

5. Conclusion & Results

- Obtained a map of Costa Rica based on the different land use classes using our Machine Learning Model.
- Built an interactive application dashboard for potential external use



6. Future Work

Adjustments & Recommendations:

- Collect feedback from Costa Rican scientists and authorities on project results and dashboard
- Update the project's pipeline and dashboard as necessary and as requested
- Enable the software and application to be available for external partners and countries

Acknowledgements

Thank you to our Corporate Partners, our professional mentors Andy Lister, John Hogland, and Rachel Riemann, as well as our peer mentor Wendy Jiang.